B. Playing with GCD

time limit per test 1 second
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 1200 | Tags: math, number theory | https://codeforces.com/problemset/problem/1736/B

You are given an integer array a of length n.

Does there exist an array b consisting of $n+1$ positive integers such that $a_i = \gcd(b_i, b_{i+1})$ for all i ($1 \le i \le n$)?

Note that gcd(x, y) denotes the greatest common divisor (GCD) of integers x and y.

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \le t \le 10^5$). Description of the test cases follows.

The first line of each test case contains an integer n ($1 \le n \le 10^5$) — the length of the array a.

The second line of each test case contains n space-separated integers $a_1, a_2, \ldots, a_n$ representing the array a ($1 \le a_i \le 10^4$).

It is guaranteed that the sum of n over all test cases does not exceed 10^5.

Output

For each test case, output "YES" if such b exists, otherwise output "NO". You can print each letter in any case (upper or lower).

Example

Input
4 1 343 2 4 2 3 4 2 4 4 1 1 1 1

Output
YES YES NO YES

Note

In the first test case, we can take b=[343,343].

In the second test case, one possibility for b is b=[12,8,6].

In the third test case, it can be proved that there does not exist any array b that fulfills all the conditions.